

MICHAL BOREK

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SUMMARY:

Software Engineer (Python/SQL) focused on **backend/platform tooling**, **test automation**, and reliable AI-assisted developer workflows. Shipped an AI-assisted SLURM submission assistant used by **~50 HPC accounts** and built reproducible ML pipelines (PyTorch/DDP). Co-author: **SPIE Medical Imaging 2025**; **Radiology: AI** (under review). Fluent in **Polish** and **English**.

TECHNICAL SKILLS

Python, C++, SQL (PostgreSQL) | Pandas, NumPy, scikit-learn, PyTorch, Captum | Flask, FastAPI, Docker, Git | Linux, SLURM, CUDA, PyTorch DDP | RAG: Qdrant, PostgreSQL FTS, AWS (SQS, S3)

EDUCATION

Michigan State University, College of Engineering (Honors College)

Expected May 2026

Bachelor of Science, Computational Data Science, Minor in Mathematics.

GPA: 3.84

EXPERIENCE

Institute for Cyber-Enabled Research (ICER), Michigan State University

East Lansing, MI

Software Engineer Intern (HPC / ML Tooling)

Jun 2023 - Present

- Built and shipped an **AI-assisted SLURM** submission assistant (**Flask service** + CLI in Open OnDemand) used by **~50 HPC accounts**; onboarded **40+** researchers in month one.
- Developed an automated **validation suite (SBATCH policy** + Linux env sanity checks + ShellCheck) with structured errors/warnings for consistent user guidance.
- Integrated a **policy-constrained LLM** limited to approved checks/outputs to reduce unsupported responses and improve **reliability**.

Institute for Quantitative Health Science & Engineering (IQ)

East Lansing, MI

Research Assistant (Medical Imaging ML)

Nov 2023 - Present

- Built a modular PyTorch pipeline for **ordinal** medical image grading with reproducible runs across data curation, training/evaluation, and automated report bundles for review.
- Owned **hyperparameter optimization** and **model selection** using held-out metrics and **calibration analysis**; executed sweeps and tracked results for consistent comparisons.
- Scaled training with **PyTorch DDP** on SLURM/CUDA, reducing wall-time from **~8h to ~3h**; improved preprocessing efficiency (e.g., Lanczos resizing) to reduce pipeline overhead.
- Produced **interpretability artifacts** with Captum (saliency/attribution) to support clinical review; contributed to **SPIE 2025** and a **Radiology: AI submission (under review)**.

Department of Computer Science at Michigan State University

East Lansing, MI

Undergraduate Teaching Assistant

Dec 2022 - Dec 2023

- Led weekly recitations for **~40** students, translating core CS concepts into hands-on problem solving and **debugging workflows**.
- Built an **auto-graded lab suite (unit tests** + rubric-based checks) that improved turnaround time by **~30%** and supported a **12%** exam performance increase.

PROJECTS

Legal Document Intelligence System (RAG)

Dec 2025 – Present

- Built an **end-to-end RAG** system for legal documents (PDF/DOCX/TXT) with citation-backed responses using hybrid retrieval (**PostgreSQL FTS** + **Qdrant**) and local Ollama inference with structured outputs.
- Implemented mandatory **verification + refusal** (claim to source validation, excerpt generation) and an asynchronous API/worker architecture with **audit logging**, retries, and safe concurrency.

*Delta Dental Capstone - **Project Lead** (Client-Facing Data Science)*

Jan 2026 - Present

- Lead partner-sponsored capstone with **Delta Dental Data Science Team**; run working sessions to align scope, milestones, and deliverables.
- Build a **canonical dataset** (stable schema + data dictionary) by consolidating source tables for consistent downstream use.
- Prototype **feature mapping to a 3D model** for interpretable visualization and consistent feature semantics.

PUBLICATIONS

- Co-author, **SPIE Medical Imaging 2025** - Ordinal pneumoconiosis grading (Computer-Aided Diagnosis)
- Co-author, **Radiology: Artificial Intelligence** - Pneumoconiosis screening (under review), 2025